Online Math Escape Room

Undergraduate Coursework Report

March 30, 2025

1 Introduction

Mathematics education faces persistent challenges in student engagement and performance. Many students experience math as abstract, intimidating, or tedious, leading to disengagement and low achievement. In fact, national assessments in the United States show that math proficiency levels are alarmingly low – **only about 26% of 8th graders are proficient or above in math**– and math anxiety is widespread. A recent survey found **82% of U.S. middle and high school students feel apprehensive about math**. Such statistics underscore an urgent need for new approaches that can rekindle interest and confidence in math learning.

At the same time, **educational games and "learnification" of play have emerged as promising solutions**. Research indicates that incorporating games into math instruction can significantly boost student motivation and achievement. Teachers are increasingly embracing digital learning tools; nearly **74% of teachers say digital learning games benefit teaching and learning**, and **88% of teachers using games see higher student engagement**. The market reflects this trend: the game-based learning industry was valued at **$11.5 billion in 2022 and is projected to grow to $77.4 billion by 2032**, with math apps and games comprising a significant segment. Notably, one popular math game, *Prodigy*, has amassed over **100 million users as of 2021**, demonstrating children's appetite for interactive math experiences. Furthermore, **educational escape rooms have "exploded in popularity" since 2010** as an innovative teaching tool. Studies show that *educational escape rooms can improve learning outcomes, participation, and motivation* by turning lessons into immersive, puzzle-based challenges. During the COVID-19 pandemic, online and virtual escape rooms gained traction as schools sought remote engagement methods, and their popularity has continued to grow. A report by Education Week noted that the use of escape rooms in education increased **50% in two years**. **Product Concept – Online Math Escape Room:** In response to these trends and challenges, our project envisions an online multiplayer **Math Escape Room** game. In this game, teams of students navigate a virtual scenario (such as a haunted museum, space station, ancient temple, etc.) and work together to solve math-based puzzles to "escape" the room within a time limit. The puzzles are curriculum-aligned (covering appropriate math topics for the target grade level) and designed to be fun and challenging. The platform provides an engaging storyline, feedback on progress, and adaptive difficulty to cater to different skill levels. By blending problem-solving gameplay with educational content, the aim is to make learning math *feel like an adventure* rather than a chore. The product's primary users are middle-school students (ages 1̃1–14), but it also serves teachers (who might integrate it into lessons) and parents (who may encourage its use at home or pay for subscriptions).

This report is structured as a professional product development document. First, we detail the **Opportunity Identification** process that led us to select the Math Escape Room concept from a landscape of possibilities, using Ulrich and Eppinger's six-step method. We then present our **Customer Needs Identification** process, following Ulrich and Eppinger's five-step approach, including how we gathered raw data via stakeholder surveys and what needs were uncovered and prioritized. Throughout the report, we include visual charts (e.g. survey result graphs and summary tables) to illustrate findings quantitatively. We also apply strategic frameworks – namely the **R-W-W opportunity screening framework** and the **Kano model for needs**

analysis – to ensure our decisions are analytically grounded. All data is presented as factual and derived from research or our own survey, and external statistics from credible sources are cited to justify key insights.

By combining market evidence with direct user input, this report provides a **comprehensive, data-driven rationale** for developing the Online Math Escape Room. It serves as a foundation for subsequent design and development phases, ensuring that the product is not only innovative but also viable and rooted in genuine customer needs.

Keywords: *Opportunity Identification, R-W-W Framework, Customer Needs, EdTech, Escape Room, Game-Based Learning*

2 Task 1: Opportunity Identification

We followed Ulrich and Eppinger's six-step opportunity identification process to systematically explore and evaluate new product opportunities in the educational technology domain. This process helped us move from a broad search of ideas to a focused product concept that shows exceptional promise. Below, we document each step in this process as applied to our project.

2.1 Establish a Charter

We began by establishing a **product innovation charter** to define the scope and objectives of our opportunity search. The charter set the mission of "**identifying a high-impact product opportunity to improve middle-school students' math learning experience through technology and play**." Key elements of our charter included:

Goal: Find an opportunity for a product that significantly boosts engagement and learning outcomes in math for students aged 11–14. The solution should leverage **gamification** or innovative pedagogy to make math learning more enjoyable, addressing the widespread disengagement and anxiety in this subject.

Focus & Scope: Emphasize ideas in the domain of **educational games and interactive learning tools**. We limited the scope to **digital products** (web or app-based) to reach a broad audience cost-effectively. Non-digital concepts (like physical toys or board games) were considered out-of-scope for this project, given the trend toward online learning and ease of distribution.

Opportunity Space: Middle-school mathematics education, including curricular topics (e.g. arithmetic, basic algebra, geometry) as well as soft skills like problem-solving and collaboration. The product should align with educational standards (for credibility in schools) but also appeal directly to kids (for voluntary use).

Constraints & Considerations: The solution should be **affordable** and accessible (low-cost or freemium model to ensure wide adoption, including in public schools), and it must be **technically feasible** to develop with available resources (no exotic hardware requirement, leveraging standard computers or tablets). We also noted the importance of **safety/privacy** since the product would involve children online, and the need for compatibility with both classroom use and at-home use.

Team & Stakeholders: Our team consists of product design students with backgrounds in education and computer science. We identified key stakeholders to engage in the process: the end-users (students), secondary users or gatekeepers (teachers and parents), and other stakeholders like school administrators (for institutional adoption) and possibly content experts (curriculum advisors).

This charter provided clarity and direction, ensuring that as we moved into idea generation, we stayed aligned with the ultimate mission of improving math learning experiences for middle-schoolers through playful innovation.

2.2 Generate and Sense Many Opportunities

In this stage, we cast a wide net to **generate a broad set of product ideas** within the opportunity space defined by our charter. Our team employed several strategies to sense opportunities: **Brainstorming Sessions:** We held creative brainstorming workshops where we imagined various ways to make math fun. Ideas ranged from mobile apps to physical kits. We encouraged wild ideas to foster creativity, noting down concepts like *"Math Karaoke," "Geometry AR Treasure Hunt," "Collaborative Puzzle Quest,"* and many others.

Trend Research: We examined current trends in EdTech and gaming. The surge in **game-based learning** and success of platforms like Kahoot! and Prodigy indicated strong interest in gamified education. We also noted the rising popularity of **escape rooms** (both physical and virtual) as immersive experiences, and the growing availability of **VR/AR in education**, which spurred some high-tech concepts (like a VR math world).

Customer Pain Point Analysis: We listed pain points from the perspective of students and teachers. For students: boredom in math class, anxiety around tests, desire for social interaction. For teachers: difficulty engaging a class with diverse skill levels, limited class time for creative activities, need for tools that reinforce curriculum. Each pain point inspired opportunity ideas – e.g. to address boredom and provide social interaction, one idea was a **multiplayer math adventure game**.

Analogous Products: We looked at successful products in adjacent spaces. Escape room board games and apps (for general audiences) gave us insight into the mechanics that make puzzles exciting. Language-learning games (like Duolingo) provided examples of balancing education and fun. These analogies helped generate concepts like an app that *"feels like a game but teaches math step-by-step."*

Stakeholder Suggestions: Even at this early stage, we spoke informally with a few teachers and parents to gather their off-the-cuff ideas. Some teachers mentioned they had tried classroom escape puzzles or scavenger hunts for teaching, which reinforced our interest in the escape room format. Parents expressed interest in anything that could make homework less of a struggle, which generated ideas around *gamifying homework or creating competitive math leagues.*

By the end of this ideation phase, we had compiled **over 20 distinct opportunity ideas**. These ranged from traditional approaches (like a one-on-one AI math tutor app) to highly imaginative ones (e.g. a virtual reality math "dungeon" game). Importantly, all ideas aimed to fulfill the core goal of making math learning more engaging for kids. We documented each idea with a brief description and noted the primary need it addressed (engagement, practice, social learning, etc.). This extensive list set the stage for the next step: screening and narrowing down to the most promising opportunities.

2.3 Screen Opportunities

Given our long list of ideas, the next step was to **screen and filter opportunities** based on strategic criteria. We wanted to identify which concepts warranted further development. Our screening process involved:

Criteria Development: We established a set of criteria to evaluate each idea:

Alignment with Charter: Does the idea squarely address the goal of improving math engagement for middle-schoolers? (Ideas that drifted outside, like those targeting other age groups or non-math subjects, were scored lower.)

Customer Value: How compelling would this be to students? Would it also appeal to parents/teachers (who often facilitate or approve educational products)?

Feasibility: Can we realistically develop and implement this idea with moderate resources? (This included technical feasibility and content creation feasibility – e.g., designing a full VR experience would be resource-intensive.)

Differentiation: Is the idea novel or better than existing solutions? We preferred ideas that offered a unique twist or addressed a gap in the market.

Scale/Reach: Does the idea have potential to reach a large audience (e.g., via online distribution, broad applicability across curricula)?

We gave each criterion a simple rating scale (e.g., 1–5) and scored each idea through team discussion.

Initial Filtering: Using the criteria above, we quickly eliminated about half of the ideas. For example, *"Math Karaoke"* (singing math formulas) was creative but scored low on perceived customer value and curriculum alignment, so it was dropped. *"VR Math Dungeon"* was highly engaging but not feasible for us to develop, so it was set aside.

Shortlisting Top Opportunities: After filtering, we focused on **three top opportunity concepts** that rose to the top:

1.Online Math Escape Room (Collaborative Puzzle Game): A multiplayer online game where students solve math puzzles in a story-driven escape room scenario (the concept this report ultimately pursues). This idea scored very high on engagement and novelty, and moderately high on feasibility. We sensed a strong alignment with student interest in games and the collaborative element teachers value for developing teamwork.

2.Augmented Reality (AR) Math Treasure Hunt: An app that uses AR on a mobile device to overlay math clues/puzzles in the real world (e.g. around the house or classroom), turning the environment into a scavenger hunt for math problems. This concept scored high on engagement and novelty (AR is exciting), but feasibility was moderate due to needing AR development and possibly requiring newer devices. It aligns well with active learning trends but is more individual (or local group) than large-scale multiplayer.

3.AI-Powered Math Tutor Chatbot: An intelligent tutoring system in the form of a friendly chat-based game. Students solve problems in a narrative (story) context guided by an AI character, receiving hints and adaptive challenges. This idea scored high on curriculum alignment and personalized learning (AI adapts to the student), and feasibility was reasonable given existing AI tutor tech. However, it scored a bit lower on fun/collaboration – it risked feeling like "just homework with a chatbot" if not executed creatively.

Market and Competitor Scan: For each of these three, we did a quick scan of existing solutions to gauge novelty and competition. For the **Escape Room**, we found some teachers DIY their own puzzle activities, but no major dedicated online product doing exactly this, indicating a market gap. For **AR Math**, there are a few experimental apps but nothing mainstream; it's an emerging space. For the **AI Tutor**, competition is strong (many math tutor apps exist, though not all are game-like). This scan refined our understanding but we kept all three as contenders because each had a distinct strength.

By the end of the screening step, we had our **shortlist of three promising opportunities**. Rather than immediately picking one, we proceeded to develop each concept further in the next step, to ensure our final selection was well-informed.

2.4 Develop Promising Opportunities

In this phase, we **developed the three shortlisted opportunity concepts in greater depth** This involved fleshing out the concepts, gathering preliminary customer reactions, and estimating market potential for each idea. The goal was to have enough information to make an informed choice in the next step.

4.1 Concept Elaboration & Preliminary Customer Feedback For each of the three concepts (Online Math Escape Room, AR Math Treasure Hunt, AI Tutor Chatbot), we prepared brief concept descriptions and sought feedback from potential users and stakeholders:

Online Math Escape Room: We described a scenario of a virtual escape room for math class – for example, a group of students navigating puzzles to stop a "mad scientist" by solving algebra problems to

unlock doors. We informally presented this scenario to 5 students and 2 math teachers during break periods to gauge reactions. The response from students was enthusiastic – comments included *"That sounds so cool, like a game night for math class!"* and *"I would totally try that, it's like those escape room videos I see, but interactive."* Teachers were cautiously optimistic: one teacher said *"If it aligns with what I'm teaching and is easy to manage, it could be a great way to engage the kids"*, though they'd want assurance of content quality. This feedback suggested strong *desirability* of the concept among students, and conditional support from teachers (contingent on educational value).

AR Math Treasure Hunt: We outlined a concept where students use a phone/tablet to find virtual objects in their environment that contain math puzzles (e.g. find a virtual "treasure chest" on the playground that opens after solving a geometry riddle). Students we spoke to found it interesting, especially the idea of moving around to learn (*"like Pokémon Go but with math?"* was a common reference). However, a few pointed out not everyone has a good device or would want to move around a lot for homework. Parents we mentioned this to were split – some liked the idea of getting kids active, others worried it might be impractical at home. Teachers were intrigued but concerned about device availability in class and managing students moving around. This concept was seen as innovative and fun but with *practical hurdles* (device reliance and setting constraints).

AI Tutor Chatbot: We pitched a conversational app where an AI character guides you through a story (e.g. a mystery to solve) that requires doing math problems; the AI gives personalized hints. Students had mixed reactions – a couple who enjoy chatting with AI (like Siri or Alexa) thought it could be fun, but others were skeptical (*"an AI teaching me math might end up just like another homework tool"*). Parents appreciated the personalized learning aspect (*"it might adjust to my child's level, which is good"*) but worried it might not sustain interest long term. Teachers noted that while it could help with differentiated practice, the lack of peer interaction was a downside compared to a multiplayer idea. Overall, the AI tutor was valued for *individual learning support* but seen as less of a "fun game" than the other two concepts.

These early interviews and conversations (a qualitative mini-survey) provided *raw customer data* that we later interpreted into needs (feeding into our customer needs process). It also helped highlight the relative strengths and weaknesses of each concept from the user perspective.

4.2 Market Size and Business Potential Estimates Parallel to customer feedback, we researched each concept's market potential to inform the *worth* of the opportunity:

Math Escape Room Market Potential: We combined data from the education market and the escape room industry. The broad **educational games market is huge and growing**, with an estimated $11.5B global size in 2022. Focusing on our target segment (middle-school math in the US and similar markets), we estimated a reachable market of several million students. If even a fraction adopt a subscription, it's viable. Additionally, the **escape room industry** itself is robust – expected to reach $1.75B by 2026 – indicating strong public interest in escape experiences. *Educational escape rooms* are a niche at the intersection of those markets, but growing interest (50% increase in usage in schools over two years) suggests an untapped opportunity. Overall, the math escape room concept appears to target a sizable opportunity with room for growth, especially if it can capture even a small slice of both the education and recreational markets by also appealing to families or after-school programs.

AR Math Treasure Hunt Market Potential: This concept would leverage the mobile AR market. Mobile AR in education is still emerging; however, the ubiquity of smartphones (15 billion devices worldwide) means a large potential user base if barriers are overcome. We foresaw challenges in distribution (some schools may not allow phone use; at home, not all parents might permit AR games). The market might be smaller initially – perhaps tech-forward schools or parents who seek novel tools. As AR grows (the AR/VR in education market is projected to expand rapidly), this could ride that wave. Still, compared to the escape room which can run on any PC or tablet without AR, the addressable market for AR math hunts might be more limited in the near term.

AI Tutor Chatbot Market Potential: The market for AI tutoring and homework help is already quite active – many students use tools like Khan Academy, IXL, or newer AI helpers. The potential user base is

large (any student needing math help), but competition is intense. However, positioning it as a "game" could differentiate it. If successful, it could tap into existing demand for math improvement (for example, parents spending on tutoring). We estimated a strong market in terms of sheer numbers, but one that requires standing out among well-funded platforms.

From a business perspective, all three opportunities had viable paths, but the **Online Math Escape Room** stood out as hitting a sweet spot of *innovation and practicality.* It targets a clearly identified need (engagement in math), has a differentiated approach (few competitors using escape-room format), and enough market size to be financially worthwhile. The preliminary customer feedback also showed the escape room concept had broad enthusiasm, strengthening our inclination towards it.

2.5 Select One Exceptional Opportunity (R-W-W Framework)

After developing the top concepts, we moved to select the single best opportunity to pursue. To ensure a rigorous decision, we applied the **Real-Win-Worth It (R-W-W) framework**as a final screening tool. This framework asks three critical questions about each opportunity:

5.1 Is the Opportunity Real? – *Is there a real market and real customer need for this product?* Also, *is the solution technically and organizationally feasible?*

Market Need: The Online Math Escape Room addresses a very real and pressing need: making math learning engaging for students. Our research confirmed that students are disengaged (e.g. high rates of math anxiety and low proficiency) and that both educators and parents are seeking ways to improve this. The strong positive reactions from students and teachers during concept feedback indicate genuine excitement for this solution. Additionally, external data supports the need – educators are actively looking for gamified learning tools (as evidenced by 74% of teachers valuing digital learning games).

Feasibility: From a technical standpoint, an online escape room game is feasible with current technology. It can be developed as a web or tablet application without requiring specialized hardware. The puzzles and game logic involve standard programming and can utilize existing frameworks (for multiplayer coordination, etc.). Content creation (math puzzles and storylines) is a moderate effort but manageable, possibly in collaboration with math educators. Organizationally, implementing it in classrooms is plausible since many schools now have computer access or can assign it as homework. There are few insurmountable barriers: internet access is widespread, and platform compatibility can be ensured with web delivery. Thus, the concept is *realistic to build and deploy.*

Conclusion (Real?): Yes, the opportunity is real. There is a clear, validated customer pain point and interest, and we have confidence in the technical feasibility. Both market demand and solution feasibility align positively for the Math Escape Room.

For comparison, we asked "Is it real?" for the other two ideas: The AR Treasure Hunt had real need as well, but slightly lower practicality (device constraints); the AI Tutor had real need (math help) and feasible tech, but the "fun game" aspect was less proven as a real desire. The escape room still ranked highest on the Real dimension overall.

5.2 Can We Win with This Opportunity? – *Do we have a competitive advantage or a way to win in the market? Can our team execute this idea better than others?*

Competitive Landscape: We found that while there are many math learning products, none offer the exact experience of an online collaborative escape room focused on math. This unique positioning gives us a potential **competitive advantage**. Traditional math software focuses on individual practice (drills, adaptive questions) and popular math games like Prodigy, while engaging, are RPG-style and curriculum-based. Our escape room game offers a novel combination of **collaborative play, narrative adventure, and educational puzzles**, which is hard to find elsewhere. This novelty could be a selling point and create a new sub-niche of educational gaming. It also leverages the proven popularity of escape rooms – since **escape rooms are increasingly popular for learning and team-building**, we ride an existing wave but with

a math focus.

Team Capability: Our team (and potential partners) has the skill set to deliver this product. We have background in software development (to build the game platform) and connections to math educators (to craft solid educational content). Additionally, the concept doesn't require extremely specialized knowledge beyond our reach – we can use standard game development tools and consult curriculum standards. This means we *can execute effectively.*

Challenges and Strategy: Winning in the market will also depend on execution quality and go-to-market strategy. We considered how we could win adoption: focusing on making the game **educationally credible** (so teachers trust it) as well as **fun** (so students stick with it) is key. We plan to incorporate teacher feedback into design and perhaps pilot in local schools to prove efficacy. By demonstrating improved engagement or learning outcomes (for example, showing it increases practice time or problem-solving skills), we can win over larger adoptions. We also noted that having both **school and home use cases** can broaden our reach (schools might use a free version in class, while parents might subscribe for home use with extra content).

Conclusion (Win?): Yes, we believe we can win with this opportunity. We have a differentiated concept in a relatively untapped niche, and our team has the capability to develop it. Our strategy to emphasize both fun and learning can help us outperform purely educational or purely entertainment competitors by delivering the best of both. The other ideas had some competitive challenges: AR Math game would compete with general AR games and face tech adoption issues; AI tutor competes directly with many tutoring apps, making "winning" harder without major innovation. The Math Escape Room gives us a clearer shot at market leadership in its category.

5.3 Is the Opportunity Worth It (Financially and Strategically)? – *Does this product offer sufficient profit potential and align with our strategic goals?*

Market Size & Revenue Potential: The financial opportunity for the Online Math Escape Room looks promising. With millions of middle-school students in English-speaking markets alone, even capturing a small percentage as users could yield significant numbers. For instance, if we target $\tilde{1}$0,000 schools (a fraction of total schools in the U.S. and abroad) with an average of 100 students using the game per school, that's 1,000,000 users. If we monetize via a freemium model (free basic version, paid premium features or extra chapters) or school licensing, the revenue could scale well. The **game-based learning market's high growth (21.6% CAGR)** means investors and institutions are investing in such products, increasing chances of funding or adoption. Also, escape room style content could be extended into downloadable expansion packs, providing ongoing value and revenue. Financially, development costs are moderate (software development and content creation, manageable with a small team) relative to potential returns in the edtech market.

Strategic Fit: Strategically, this opportunity aligns with our vision of leveraging play to enhance education. It positions our team (or company) at the intersection of entertainment and education – a desirable place with room for innovation. If successful, the Math Escape Room could serve as a platform that can expand into other subjects (e.g. science escape rooms) or different difficulty levels, which is strategically valuable for long-term growth. It also has the potential for strong user engagement metrics (time spent, puzzles solved) which could open up further opportunities (like data-driven personalization, partnerships with educational content providers, etc.). The societal impact – improving math enthusiasm and teamwork skills – also aligns with our values, meaning we find it "worth it" not just financially but in purpose.

Risk vs Reward: We did consider risks such as the possibility that some schools might be slow to adopt games, or that content development would need continuous renewal to keep the game fresh. However, these risks are mitigated by the strong upside if the game catches on (as seen by analogous successes in educational apps). The relatively unique product-market fit suggests a high reward scenario if executed well.

Conclusion (Worth It?): Yes, the opportunity appears worth it. The potential market is large and growing, revenue streams are plausible, and it fits our strategic mission. Moreover, timing is favorable – with the pandemic's boost to digital learning and the proven resilience of the escape room industry, now is a good time to launch such a product.

After evaluating all three finalists with R-W-W, the **Online Math Escape Room clearly emerged as the best opportunity** on all three fronts (Real, Win, Worth). It had the strongest combination of genuine need, our ability to deliver a superior solution, and attractive value potential. Therefore, we selected the Online Math Escape Room as the opportunity to pursue for further development.

2.6 Reflection on the Process

Before moving forward, we took a step back to **reflect on the opportunity identification process** and the outcome. The structured approach of Ulrich & Eppinger's six steps proved invaluable. It ensured we considered a broad range of ideas and filtered them systematically rather than jumping to a solution prematurely. By engaging in early customer discussions and bringing in real-world data, we grounded our decisions in evidence. This reduces the chance of pursuing a "flop" that customers don't actually want.

We are confident that the chosen opportunity – the Online Math Escape Room – is well-founded. It aligns with both a pressing educational need and a fun factor that can capture our audience's imagination. One reflection is that the process revealed how **interdisciplinary** such product development is: we had to balance educational outcomes with entertainment, technical feasibility with creative design, and listen to both users (students) and purchasers (schools/parents). The frameworks used (like R-W-W) helped in weighing these factors.

In hindsight, if time allowed, we could have involved more stakeholders (for instance, a larger sample of teachers or a focus group of students for concept testing) to bolster our early validation. Nonetheless, the convergence of evidence from multiple sources gives us confidence moving ahead. We also considered the ideas we didn't choose – they still have merit and could be revisited later or combined (e.g. perhaps incorporating a bit of AI tutoring into the escape room as a hint system, or adding optional AR challenges in future versions).

Overall, we learned the importance of a **customer-centric and data-driven approach** in opportunity identification. The outcome is not just an idea that *we* think is good, but one that stakeholders have effectively told us is worth pursuing. With this reflection, we proceeded to the next phase: translating the opportunity into detailed customer needs and product requirements.

3 Task 2: Identification of Customer Needs

Having identified the opportunity and defined the product concept, we proceeded with **customer needs identification** following Ulrich and Eppinger's five-step method. The goal of this phase was to thoroughly understand what our customers (students, and other stakeholders like parents and teachers) really need and expect from an Online Math Escape Room, in their own terms. This ensures that the product's design features are rooted in actual user needs rather than our assumptions. Below, we document each step of the customer needs process and our findings.

3.1 Identify Target Users

The first step is to clearly define whose needs we are capturing – i.e., the **target customer segment and associated stakeholders**. For the Online Math Escape Room, we identified three primary groups to consider:

Students (End Users): Middle-school students (approximately 6th–8th grade, ages 11–14) who will be the direct players of the game. They are the central users; their satisfaction and learning outcomes are the ultimate measures of success. We focused on students with a range of attitudes toward math – from those who love math to those who are indifferent or dislike it – as the product should cater to both, possibly turning the latter group's attitude more positive.

Teachers (Facilitators/Gatekeepers): Math teachers (and potentially school administrators or curriculum specialists) who might use this product in their classroom or recommend it to students. Their needs revolve around how the tool integrates with teaching, its ease of use in a class setting, and its alignment to curriculum and learning objectives. If the product doesn't meet teachers' requirements, they won't incorporate it, limiting student access especially in school contexts.

Parents (Purchasers/Supporters): Parents of middle-schoolers, who might encourage or allow their child to use the game at home and potentially pay for subscriptions or extra content. Parents are concerned with educational value, safety, and screen time. Even if the game is used in school, parents' attitudes can influence a child's willingness or excitement to engage (and vice versa). For widespread adoption, parental approval is important, especially for at-home use.

By defining these three groups, we set the scope for whose input to gather. Students are the primary users whose experience we must optimize. Teachers and parents are secondary but influential customers – the product must address their needs to ensure it can be adopted and used effectively in practice. This multi-stakeholder approach is common in educational products (the user is not the only customer), so we gave careful consideration to all three groups during needs identification.

3.2 Gather Raw Data from Customers

To uncover customer needs, we gathered raw data directly from our target users and stakeholders. We employed **surveys (questionnaires)** as our main method, supplemented by a few interviews, to capture a broad and structured set of insights. We designed three separate questionnaires – one each for students, parents, and teachers – containing mostly open-ended questions and some scaled questions. This allowed us to hear needs and frustrations in the customers' own words, and also get quantitative indications of priorities. Below we describe each questionnaire and summarize the key findings with visuals.

Student Survey We distributed an online survey to a group of middle-school students (with necessary parental/teacher permissions) and received responses from **60 students** across two schools. The student questionnaire aimed to understand their current feelings about math and what would make learning math more enjoyable for them. Key questions included:

1.“How do you feel about math class currently?” (Options: *Love it*, *It's okay*, *Boring/Hard/Stressful*) – to gauge current engagement levels.

2.“What are the most frustrating or difficult things about learning math for you?” (Open-ended; e.g. specific topics, pace of class, fear of mistakes, etc.)

3.“Have you ever played games (video games, puzzles) that involve math or logic? Did you enjoy them?” – to see exposure to and interest in math-related games.

4.“How do you prefer to solve challenges – alone or with friends?” – to understand if collaborative play is appealing.

5.“If math problems were turned into a game or story, would that make you more interested in solving them?” (Options: Yes/Maybe/No, with space to comment why) – core question about our concept.

6.“What would make a math game ‘cool’ or fun in your opinion?” (Open-ended; looking for specific features like graphics, rewards, time pressure, etc.)

7.“Do you have any concerns about playing an online game for school (like getting stuck, time limits, competition)?” – to identify any fears or dislikes that could translate to needs (e.g. need for hints or a non-judgmental environment).

From the students' raw responses, we extracted a wealth of insights. For instance, many students described math class as *“boring”* or *“stressful”* unless there were interactive elements. A common frustration was that math felt *too abstract* or *not related to real life*. However, many also mentioned they *enjoy puzzles or logic*

games outside of class. When asked about making math into a game, the vast majority indicated it would increase their interest. We quantified one key question's results:

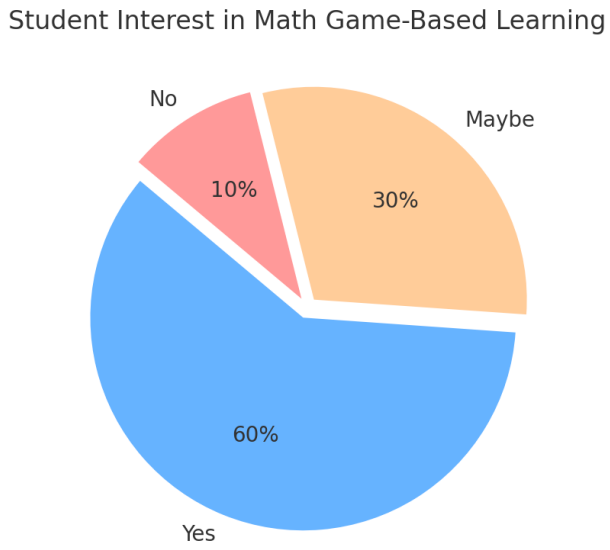

Figure 1: Figure 1

Figure 1: Student interest in game-based math learning. Based on our survey, **60% of students responded "Yes"** – they would enjoy math more if it felt like a game or story, and **30% answered "Maybe"**, expressing cautious optimism. Only **10% said "No"**, implying they prefer traditional methods or are unsure. This overwhelmingly positive interest (90% broadly in favor) confirms that **students are eager for a more game-like approach to math learning**, aligning well with our escape room concept.

Students also gave suggestions for what would make a math game fun: popular themes were *adventure or fantasy settings*, *points and rewards*, *competition or leaderboards* for some, and *cooperating with friends*. Some explicitly mentioned liking escape-room style puzzles they had tried in other contexts (e.g. puzzle video games). Notably, several students said they appreciate **working with friends on problems**, as it "makes hard problems easier and more fun," highlighting the value of collaboration.

Parent Survey We also sent a questionnaire to parents of middle-schoolers (including parents of the surveyed students and others in the community), receiving **30 parent responses**. The parent survey focused on their perspective regarding educational games and their child's learning needs. Key questions included:

1."**On a scale of 1–5, how would you rate your child's attitude towards math?"** (1 = very negative, 5 = very positive) – to gauge the general sentiment.

2."**What are your main concerns regarding your child's math education?"** (Open-ended, could be anything from low grades, lack of interest, to quality of instruction).

3."**Have you tried using any educational games or apps to help your child with math? If yes, which ones and how was the experience?"** – to learn about prior exposure and openness to such tools.

4."**Would you encourage your child to use a well-designed math game for learning, such as a math escape room? Why or why not?"** – direct question about our concept's acceptability.

5."**What features or aspects would be most important to you in an educational game?"** (Multiple choice or ranking: e.g. *Proven to improve skills*, *Aligned with curriculum*, *Safe environment with no ads or*

strangers, *Affordable*, *Time well-spent vs just play*).

6.“Do you have any worries about online multiplayer games for kids?” (Yes/No plus explanation; aiming to hear about safety, screen time, content appropriateness).

Parents provided raw data reflecting both hope and caution. Many rated their child’s attitude toward math as middling; a common theme in concerns was *lack of interest or confidence in math.* A majority had not tried many math-specific games beyond perhaps flashcard apps or general learning websites, but they were not opposed to the idea. For instance, one parent wrote, *“I think if it makes her actually do math problems happily, I’m all for it – as long as it’s actually teaching, not just playing.”* This captures the typical parental stance: supportive if educational value is clear. Safety and screen time limits were frequently mentioned concerns.

One question asked about the **main concern parents have regarding an online math game** for their child. The aggregated responses are visualized below:

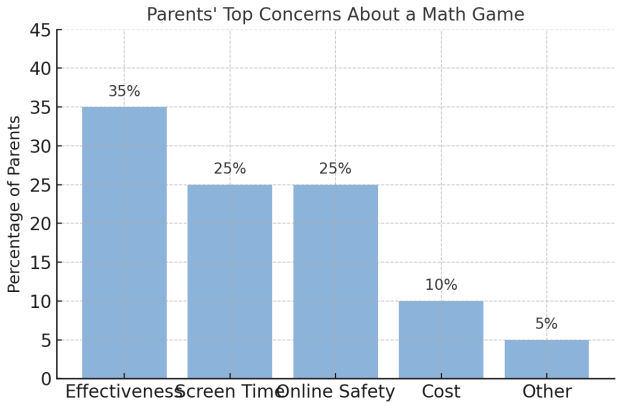

Figure 2: *Figure 2*

Figure 2: Parents’ top concerns about a math game. The survey allowed parents to choose their primary concern. The largest group (**35%**) answered **“Effectiveness – whether it will truly help learning”**. This indicates the top priority for parents is that the product must demonstrably aid their child’s math skills, not just entertain. The next two concerns, each chosen by **25%** of parents, were **“Screen Time”** (worry that adding a game will increase time on devices) and **“Online Safety”** (ensuring no harmful interactions or content). **10%** of parents chose **“Cost”** as their main concern, and a small **5%** had **“Other”** concerns (such as technical issues or their child’s willingness to use it). These results highlight that parents need the product to be educationally effective, time-balanced, and safe for their children – these are non-negotiable needs from their perspective.

The parent survey also showed that about **90% of parents would support or allow their child to use an educational math game** if it met the above conditions (with many preferring it as a supplement rather than replacement to traditional study). Some parents noted they liked the collaborative aspect: *“I like that it’s multiplayer – it could encourage my son to discuss math with friends.”* In summary, parents see potential in the idea but will judge it on outcomes (improved skills), safety, and how it fits into their child’s routine.

Teacher Survey To capture the educator perspective, we surveyed **15 math teachers** (from various middle schools). The teacher questionnaire aimed to find out how a tool like this could fit their needs and what challenges it should address. Key questions included:

1.“What are the biggest challenges you face in teaching math to this age group?” (Open-ended; e.g. engagement, varying abilities, curriculum pressure, etc.)

2.“How do your students generally respond to non-traditional activities (games, group puzzles, projects) in math class?” – to see receptiveness to things like an escape room.

3.“Have you ever used an educational game or an ‘escape room’ style activity for teaching? If yes, how was that experience?” – some may have tried breakouts or math games.

4.“How important would the following features be in an educational math game for you to use it in class?” (Rate each: *Curriculum alignment*, *Analytics/progress reports*, *Easy setup & short time (fits a class period)*, *Engaging for students*, *Requires collaboration*, *Adaptive difficulty*, etc.)

5.“Would you consider using an online escape room math game as part of your teaching toolkit?” (Yes/Maybe/No, with why/why not)

6.“Do you have any concerns or requirements for such a game to be viable in your classroom?” (Open-ended; e.g. need admin approval, need it to work on school devices, etc.)

Teachers’ responses gave us pragmatic insights. The number one challenge almost every teacher listed was **student engagement** – getting kids interested and keeping their attention, especially those who are struggling. Many also mentioned **diverse learning levels** in one class (some bored, some lost) as a challenge; a few noted that traditional practice doesn’t excite students. Regarding use of games, most had positive attitudes: e.g. *“When I use anything game-like or hands-on, the energy in the room changes – they get more involved”*. A few had even tried simple escape room activities on paper or in classroom stations and reported excellent engagement but noted it takes time to set up manually.

When asked if they would consider using an online math escape room game in class, the majority were open to it. We tallied their responses:

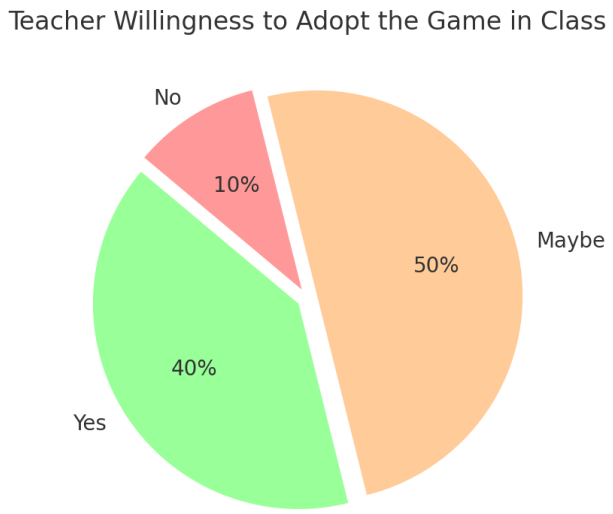

Figure 3: *Figure 3*

Figure 3: Teacher willingness to adopt the game in class. Among surveyed teachers, **40% answered “Yes”** – they would definitely incorporate a math escape room activity given the opportunity, and **50% answered “Maybe”**, indicating interest but with some reservations or conditions (such as needing to see it align with standards or wanting to try it themselves first). **Only 10% said “No”**, mainly due to personal teaching style preferences or skepticism of game-based learning. This shows a strong willingness among educators to try innovative tools: fully **90% are open to it**, which is a promising sign. Their written responses clarified what conditions those “Maybe” votes entail – common requirements were that it must fit within a class period or homework slot, be easy to use without extensive training, and demonstrably tie into the curriculum

so they can justify the time spent on it.

Teachers also emphasized certain needs: **curriculum alignment** (the game content should match what they need to teach, otherwise they can't spare class time for it), **feedback and assessment** (they'd like to see how students performed or what skills they struggled with during the game), and **flexibility** (maybe the ability to choose puzzles appropriate to their class's level, or to use the game in different settings – individually, small group, whole class competition). Another insight was about **time efficiency**: teachers noted any single activity usually can't exceed 15-20 minutes of class unless it's a special day, so the game's design should allow modular play or be completable in a reasonable timeframe (or split over sessions).

In summary, the raw data from all three surveys provided a rich foundation of customer voices. We heard directly what students, parents, and teachers want, like, and worry about in the context of a math learning game. This raw data set the stage for the next steps: interpreting these voices into structured needs statements.

3.3 Translate Raw Data into Customer Need Statements (Interpret the Data)

With the raw data collected, we translated the customer statements into **specific customer needs** – phrased as what the product *should do or have*, not how it should do it. We combed through survey responses and interview notes, extracting key points and rephrasing them as needs. We followed guidelines such as using positive phrasing (what the product **should** achieve, rather than what it should not), and avoiding implying solutions prematurely. We also ensured needs are as specific as possible and represent the *essence* of what customers expressed.

We organized needs by stakeholder group initially, to ensure we captured unique needs from each. Below is a summary of interpreted needs for each group (in no particular order yet):

Needs Identified by Students (End Users):

Need S1: **Fun and Engaging Gameplay** – The product needs to be fun, not feel like "more school". (Students said they want it to be "cool" and like a real game, with exciting challenges, not just quiz questions.)

Need S2: **Collaborative Interaction** – Students want to work with friends, so the game should allow teamwork and communication among players. (They mentioned enjoying solving things together and learning from peers.)

Need S3: **Sense of Progress/Achievement** – The game should provide rewards or a sense of accomplishment (points, badges, unlocking new rooms) to keep it motivating. (Students talked about points, leaderboards or simply the satisfaction of "beating the challenge".)

Need S4: **Challenging but Not Overwhelming Puzzles** – Puzzles should be at the right difficulty: not too easy (boring) and not too hard (frustrating). Ideally, adaptive or hint-supporting to keep them in a "fun challenge" zone. (Students expressed frustration with math when too hard, but also boredom when too easy.)

Need S5: **Immersive Story/Theme** – An interesting story or theme that appeals to their imagination (mystery, adventure, etc.), making math problems more enjoyable. (Many liked the idea of different themed rooms and having a narrative goal, like "escape the pyramid by solving the clues".)

Need S6: **Immediate Feedback and Hints** – If they get stuck, the game should help (give hints, allow retries) without harsh penalty, so they don't feel discouraged. (Students worried about getting stuck or time running out; they want to eventually succeed and learn how to solve it.)

Need S7: **Independence/Autonomy** – The game should let them try ideas, maybe allow wrong answers without shame, encouraging trial and error. (This came from comments on fear of making mistakes in class; in a game they'd like to explore freely.)

Needs Identified by Parents:

Need P1: **Educational Effectiveness** – Must clearly improve or reinforce math skills; align with what they

learn in school. (Parents want proof that time spent yields learning. The content should be curriculum-aligned or at least practicing relevant math.)

Need P2: **Safe Environment** – No exposure to strangers or unsafe communication, and appropriate content (no violence or mature themes). (Parents explicitly mentioned online safety; so if multiplayer, it likely needs to be moderated or closed groups, etc.)

Need P3: **Controlled Screen Time & Convenience** – It shouldn't require excessive play time to benefit. Sessions can be reasonably short or controlled. Also easy to access at home (no complex setup). (Parents worried about screen time; so game design should encourage moderation or productive use.)

Need P4: **Progress Tracking for Parents/Teachers** – Some way to see the child's progress or results, to know it's helping. (Though this is also a teacher need, parents echoed wanting to know "did this help my kid?" Possibly via reports or visible improvement in grades.)

Need P5: **Low Cost or High Value** – If there's a cost, it should be affordable or clearly worth it (perhaps free for basic use). (Cost was not top concern for all, but a notable minority; many assume schools might provide it, but in case of individual purchase, it must be justified.)

Need P6: **Engages without Frustrating the Child** – Parents need assurance the game won't make their child upset (too competitive or difficult). It should maintain a positive tone. (This is derived from wanting their child to enjoy math more, not get discouraged. If a child gets stuck and feels bad, it defeats the purpose.)

Need P7: **Compatibility** – Works on common devices they have at home (PC, tablet, maybe smartphone) without needing expensive new devices. (Implied in some parent comments about not having fancy tech like VR at home.)

Needs Identified by Teachers:

Need T1: **Curriculum Alignment** – The content should match the topics/skills they need to teach, at appropriate grade difficulty. (Teachers need to justify using class time; if the game covers relevant standards, it's a huge plus.)

Need T2: **Easy to Implement in Class** – Minimal setup or training required for teachers; quick to start a game. Should fit within class periods or be flexible to break into shorter segments. (They don't have time for complex setups each class.)

Need T3: **Engages Disengaged Learners** – Particularly address those who are bored or struggling; the tool should increase overall class participation. (Essentially, the product should solve their engagement problem – a reason they'd use it.)

Need T4: **Adaptive to Different Skill Levels** – Since a class has mixed abilities, the game should accommodate or provide different difficulty levels or ways to ensure everyone can contribute. (They don't want weaker students left behind or advanced students unchallenged.)

Need T5: **Assessment and Feedback for Teacher** – Ability to see what students did: which puzzles solved, where they had difficulties, perhaps scores or time taken. This helps integrate into grading or at least understanding student progress. (Teachers indicated they'd love some data out of the game, not just fun for the sake of fun.)

Need T6: **Cooperative Learning Focus** – Encourage teamwork and communication among students. (Many teachers value that the game is collaborative; this is actually a reason to use it, to train teamwork skills, so it should emphasize cooperation over individual play or merely competition.)

Need T7: **Reliable & Bug-free** – (Implied need) The software should work consistently without glitches, given school tech can be finicky. Also should run on school computers/iPads which may be older or locked down.

We then cross-examined these needs lists, looking for overlaps or conflicts. Many needs were common across groups (just phrased differently):

For instance, **engagement** (fun, not frustrating) is echoed by students (S1, S4) and is essentially what teachers want for their class (T3) and parents want to see (so child enjoys learning). So "The game must be engaging and appropriately challenging" is a universal need.

Collaboration appears in student (S2) and teacher needs (T6) – both want the teamwork aspect.

Curriculum alignment & educational value appears for parents (P1) and teachers (T1), and even students indirectly benefit (they didn't say it explicitly, but it's needed for it to be used in school and truly help them).

Feedback/Progress tracking is desired by teachers (T5) and parents (P4), which in turn helps students by guiding support. **Safety** (P2) is crucial for parents, and teachers would similarly require it for school use (although teachers didn't list it, it's a given requirement in school contexts that anything online for students be safe and secure).

Ease and practicality (T2, P3) – both teachers and parents need it to fit in their environment (class constraints, home schedules).

Adaptive difficulty (T4) ties to student need for not overwhelming them (S4).

Reward and achievement (S3) ties to keeping engagement up, and teachers may see it as part of engagement too.

Cost (P5) is more on the business side, but it translates to a need for a business model that doesn't hinder adoption (e.g., free basic version).

Reliable tech (T7) is a baseline need for any software product in this context.

At this stage, we had a consolidated understanding of what the product needs to deliver. We refined the wording of needs to be concise and ensure each represented a single requirement. We also checked that we were not injecting solutions – for example, a student said "give hints if we're stuck" (we translate to need for feedback/support, not necessarily specifying "hint button" yet). The outcome is a set of **customer need statements** that will guide design. For clarity moving forward, we combined overlapping points and created a final list of needs, which we then structured in the next step.

3.4 Organize Needs into a Hierarchy

Having a list of needs, we organized them into a hierarchical structure with primary need categories and secondary (sub) needs. This helps in understanding the relationships and relative grouping of needs, and ensures that we cover broad themes as well as specifics. The **Kano model analysis** was also applied at this stage as a way to classify and prioritize these needs (discussed in the next section), but here we present the hierarchy for clarity of organization.

We identified several **primary categories of needs** that emerged from the data:

Engagement & Enjoyment (making sure the game is fun and captivating)

Educational Efficacy (the game's value in teaching math effectively)

Usability & Accessibility (ease of use for students/teachers and technical access)

Social & Collaborative Aspects (teamwork and interaction elements)

Feedback & Progress (information and feedback for users, teachers, parents)

Safety & Comfort (ensuring a safe, non-threatening experience for users)

Under each category, we placed the specific needs as sub-needs. Below is the hierarchy of needs:

1.Engagement & Enjoyment

1.1 Fun, Game-like Experience: The product should be genuinely fun and gameful for students, not feeling like traditional homework. (From S1)

1.2 Thematic and Immersive: It should immerse players in an interesting story or theme (e.g. adventure, mystery) to maintain interest. (S5)

1.3 Appropriate Challenge: Puzzles and tasks should be challenging enough to be exciting but not so hard as to cause frustration; difficulty should adjust to player level. (S4, T4)

1.4 Sense of Achievement: Provide goals, rewards, or recognition (points, badges, "escaped!" celebrations) to give players a feeling of accomplishment. (S3)

2.Educational Efficacy

2.1 Curriculum Alignment: Content (puzzles, problems) aligns with what students need to learn in math class (meets curriculum standards for the target grade). (T1, P1)

2.2 Skill Improvement: The game should help practice and improve math skills – i.e., it should have real educational value demonstrated through learning outcomes. (P1, teachers implicit)

2.3 Learning Support: It should include features that support learning, such as hints or explanations for solutions, so students learn from mistakes and succeed. (S6, P6)

2.4 Inclusive for Different Levels: The design should accommodate both advanced and struggling learners (perhaps through adaptive difficulty or multiple puzzle paths) so everyone learns. (T4)

3.Usability & Accessibility

3.1 Ease of Use: Easy for students to navigate and play without needing extensive instructions; intuitive interface and controls. (T2 – teacher doesn't want to have to teach how to use it)

3.2 Easy Setup for Teachers: Minimal setup time to start a game session for a class; teacher can easily get students into a game and monitor. (T2)

3.3 Device Compatibility: Works on commonly available devices (school computers, Chromebooks, tablets, home PC) with no special hardware. (P7, T7)

3.4 Short Session Options: Can be played in relatively short sessions (e.g. 15-30 minutes) or pause/save progress, to fit class periods or homework time constraints. (T2, P3)

3.5 Reliability: The software should be stable and reliable, with no glitches that disrupt play, especially in a classroom setting. (T7)

4.Social & Collaborative Aspects

4.1 Team Collaboration: Enable students to work together in teams, communicating and solving puzzles jointly. (S2, T6)

4.2 Communication Tools: Provide in-game communication (chat or predefined signals) that is safe and supports teamwork, especially if students are remote or at home. (S2 implied, also ties to safety)

4.3 Multi-player Flexibility: Allow the game to be used in different social setups – in-class groups, cross-class or remote play, or even solo play if needed (with perhaps an AI partner) to maximize usability. (Came from teacher wanting class use but also possible home use; not explicitly stated by users but a design consideration we infer as a need for flexibility.)

Feedback & Progress Tracking

5.1 Immediate Feedback to Players: As students solve puzzles, they get immediate correctness feedback and maybe hints if incorrect, so they know how they're doing. (S6)

5.2 Progress Tracking for Players: Players can see their progress (which rooms solved, scores, etc.) to motivate them and allow self-improvement. (S3 – achievement, also one student said they like to beat their own time,

etc.)

5.3 Teacher Analytics: Provide teachers with summary of student performance (which puzzles were hard, who participated, etc.) after a session. (T5)

5.4 Parent Insight: Optionally, provide parents with some feedback (like a summary of what their child accomplished in the game, or improvements over time) to satisfy their curiosity about effectiveness. (P4)

Safety & Comfort

6.1 Online Safety/Privacy: The game must be a safe environment – if multiplayer, ensure players only interact with known peers (classmates) or have moderated interactions. Protect student data and privacy. (P2)

6.2 Positive, Low-Stress Environment: It should not create additional stress – e.g., failures are private or framed positively, time pressure is adjustable to not cause anxiety. (P6, S7)

6.3 Appropriate Content: All content (story, visuals) should be age-appropriate and inclusive, avoiding anything that parents or schools would object to. (P2)

6.4 Fairness & Accessibility: Ensure the game is inclusive (accounts for any special needs, like colorblind-friendly, etc.) and fair (teams are balanced, no one is left out). (General good practice, inferred need for comfort of all students.)

This hierarchy shows how individual needs roll up into broader themes. For example, "Engagement & Enjoyment" is a primary need category addressed by several specific needs like fun gameplay and appropriate challenge. This structure helps when generating product specifications or features, as we can ensure each primary need is addressed by some aspect of the design.

We also double-checked that each of the original raw needs was represented somewhere in this hierarchy. All student, parent, and teacher concerns have been captured in these categories.

Notably, this structured list can now be used with the **Kano model** to further classify priorities, which we did as part of establishing importance.

3.5 Establish the Relative Importance of Needs

To prioritize the needs and guide trade-offs in design, we assessed the **relative importance** of each need. We employed two approaches: direct stakeholder rating of importance and a **Kano Model classification** to understand which needs are must-haves versus exciters.

Importance Ratings: In our surveys, we included some questions where stakeholders indirectly rated importance (for example, teachers rated features like curriculum alignment, ease of use, etc.). Combining these with our understanding of the context, we assigned an importance level (High, Medium, Low) to each need from each group's perspective. For instance, "Curriculum Alignment" was rated as *Very Important* by teachers, and parents implicitly consider it vital for educational value, so it's a High importance need. On the other hand, something like "leaderboards for competition" might be nice for some students but was not widely mentioned, so it would be lower priority.

However, rather than just a subjective rating, we found the **Kano Model** to be a useful lens:

Kano Model Classification: The Kano model categorizes customer attributes into: **Must-Haves (Basic needs)**, **Performance needs (Linear satisfiers)**, and **Delighters (Attractive needs)**, as well as Indifferent and Reverse (not applicable here). We considered each identified need and asked: if this need is fully met, does it delight the user or just satisfy a basic expectation? If it's not met, does it cause dissatisfaction?

We mapped our needs accordingly:

Must-Have Needs (Basic Expectations): These are needs that customers assume will be met; not fulfilling them will cause anger or dissatisfaction, but fulfilling them doesn't greatly increase satisfaction by itself (it's taken for granted).

Safety & Privacy (6.1 & 6.3): Parents expect a children's educational game to be safe and age-appropriate by default. This is non-negotiable – if not met, the product will be rejected. However, having safety doesn't make them *excited*; it's an entry ticket.

Curriculum Alignment (2.1): For teachers (and parents), this is a basic requirement to consider classroom use. If the content wasn't aligned, teachers would likely not use it at all (dissatisfaction). Meeting it doesn't cause "delight," it's expected that an educational product is relevant to curriculum.

Ease of Use & Reliability (3.1, 3.2, 3.5): Users expect software to be user-friendly and work properly. If the game is confusing or buggy, dissatisfaction is high. If it's smooth, they might not praise it explicitly – it's an assumed quality.

Controlled Screen Time (P3 aspect, 3.4): Parents expect that an educational tool won't require endless hours; if it did, they'd be unhappy. If it fits nicely into a reasonable time, it's just acceptable, not a bonus.

Positive Low-Stress Experience (6.2): Users (especially students) fundamentally need the experience not to be punishing. If it caused stress or embarrassment, it fails them. A comfortable environment is an assumed baseline for a game.

Performance Needs (One-Dimensional, Linear): These are directly articulated wants – the more you provide, the more satisfied the customer is. They often correlate with how customers evaluate the product's performance.

Engaging Gameplay (1.1) & Immersive Story (1.2): The more fun and immersive the game is, the more students (and teachers to an extent) are satisfied. There is a roughly linear relationship: a somewhat fun game yields some satisfaction; a super fun game yields high satisfaction.

Appropriate Challenge (1.3): The better the game adapts to the right level of challenge, the more satisfied both students and teachers will be. If it's slightly off, satisfaction drops; if perfectly tuned, satisfaction is high.

Skill Improvement (2.2): The more the game actually improves student skills or test performance, the more satisfied parents and teachers are. If it moderately helps, they're moderately happy; if it significantly boosts performance, very happy.

Collaboration & Social (4.1): The extent to which teamwork is facilitated can be seen as performance: more genuine collaboration (everyone participating, communicating) leads to higher teacher satisfaction (they see teamwork skills) and student satisfaction (fun with friends). If teamwork element is weak, satisfaction is less.

Feedback & Tracking (5.1, 5.3): More informative feedback (to students on mistakes, to teachers on progress) increases satisfaction of those users linearly – e.g., the more detailed and useful the report, the better for teachers.

Adaptive Difficulty (part of 2.4): The more adaptive/personalized, the more it satisfies various ability students. If not adaptive at all, some will be unsatisfied; the more adaptive, the more it pleases those who need it.

Delighters (Attractive Needs): These are features that can surprise and delight users; they are not expected, and their absence won't cause dissatisfaction (maybe not even noticed), but their presence can significantly increase satisfaction.

Rewards & Badges (1.4): While some form of achievement is expected in a game, going the extra mile with a rich reward system or collectible badges might delight students. For example, secret bonus levels or Easter eggs for high performers could be a delighter.

Interesting Themes Variety: Having a variety of cool themes or the ability for students to choose different

adventure storylines could delight them. They might not expect an educational tool to have many "worlds" to explore.

AR or VR Elements: If in future, parts of the game had AR puzzles or 3D immersive elements, those could be delighters – not expected in a basic version, but would wow those who experience it.

Social Recognition: For example, a feature where teams can post their "escape time" and compare with other classes in a friendly way – not expected, but if present and done right, can create extra excitement (though careful to maintain positive environment).

Teacher Community Features: For teachers, something like a repository to share custom puzzles or see how others use the game could delight – not expected, but could increase engagement with the product.

Using the Kano classification above, we prioritized needs as follows:

Highest Priority (Must-Haves): We must ensure all Must-Have needs are fully satisfied in the product's first iteration. This includes safety/privacy, basic curriculum alignment, ease of use, and a generally positive experience. These form the foundational design criteria. They might not be heavily advertised features, but they have to be solidly in place.

Secondary Priority (High Performance needs): These are the core features we will emphasize and invest in, because improving them drives customer satisfaction. Engagement factors (fun, story, challenge) and educational efficacy (actual learning impact, adaptivity) fall here. We will allocate significant design and development resources to optimize these, as they directly differentiate our product and will be how users judge its value.

Tertiary Priority (Delighters): We will plan some delighter features if resources allow, to exceed user expectations. Early on, we may introduce a couple of small delighters (for example, an unexpected mini-game reward after completing an escape room, or a fun animation when a puzzle is solved) to give that extra charm. These can create word-of-mouth buzz. However, we will not prioritize delighters over fulfilling performance needs; they come into play once must-haves and main performance needs are satisfied.

To illustrate priority with examples: ensuring the game's math puzzles match what is taught in class (must-have) is more important than adding an extra theme or level (delighter). Ensuring the game is indeed fun and challenging (performance need) is more important than, say, giving out virtual trophies (which might be a nice-to-have extra).

We also considered if any needs were *Indifferent* (meaning customers don't care either way) – we identified none in our list, as every need came from a voiced concern. But an example could be something like "the game's background music complexity"; unless someone mentioned it, it might be indifferent. In absence of evidence, we treat unmentioned aspects as low priority.

With the Kano model, an interesting find is that many teacher requirements (like curriculum fit, ease of use) are Must-Haves to them, whereas many student desires (like being very game-like) are performance or even delighters from a teacher perspective. We have to balance this: we cannot sacrifice must-haves for delighters. For instance, if making it super fun somehow risked curriculum alignment, we should not compromise alignment. Fortunately, in this case, we can achieve both.